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Exercise No: 3

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EXTRACTION OF WATER BODIES AND RIVERS USING QGIS

Objective: To create a false color composite from satellite images and extract water bodies and rivers using shapefiles in QGIS.

Software Required:

- QGIS (latest version installed)
- Four satellite image files in **.tif** format

Procedure:

1. Add Raster Files

- Open Data Source Manager → Raster → Browse → Select four **.tif** files → Add → Close.

2. Create Virtual Raster (Combine Files)

- Go to Raster tab → Miscellaneous → Build Virtual Raster.
- Input Layers: Select the four **.tif** files.
- Check “Place each input file into a separate band”.
- Save Output → Select Location → Run → Close.

3. Generate False Color Composite

- Right-click on the combined raster → Properties → Symbology.
- Assign: Red = Band 4, Green = Band 3, Blue = Band 2.
- Apply → OK → Close.

4. Extract Water Body (Polygon Shapefile)

- Go to Layer tab → Create Layer → New Shapefile Layer.
- File Name: **polygon** → Geometry Type: Polygon.
- Add Fields:
 - Name: String (80)
 - Type: String (80)
- OK.
- Right-click on polygon layer → Toggle Editing.
- Use Add Polygon Feature tool → Select the water body region.

- Enter Attributes: ID = 1, Name = A, Type = Water Body → OK.
- Right-click on polygon layer → Save Layer Edits.

5. Extract River (Line Shapefile)

- Go to Layer tab → Create Layer → New Shapefile Layer.
- File Name: **line** → Geometry Type: Line String.
- Add Fields:
 - Name: String (80)
 - Type: String (80)
- OK.
- Right-click on line layer → Toggle Editing.
- Use Add Line Feature tool → Digitize the river.
- Enter Attributes: ID = 1, Name = A, Type = River → OK.
- Right-click on line layer → Save Layer Edits.

6. Style the River Layer

- Right-click on line layer → Properties → Symbology.
- Change Color: Yellow–Green mix.
- Set Line Width: 0.5 mm.
- Apply → Close.

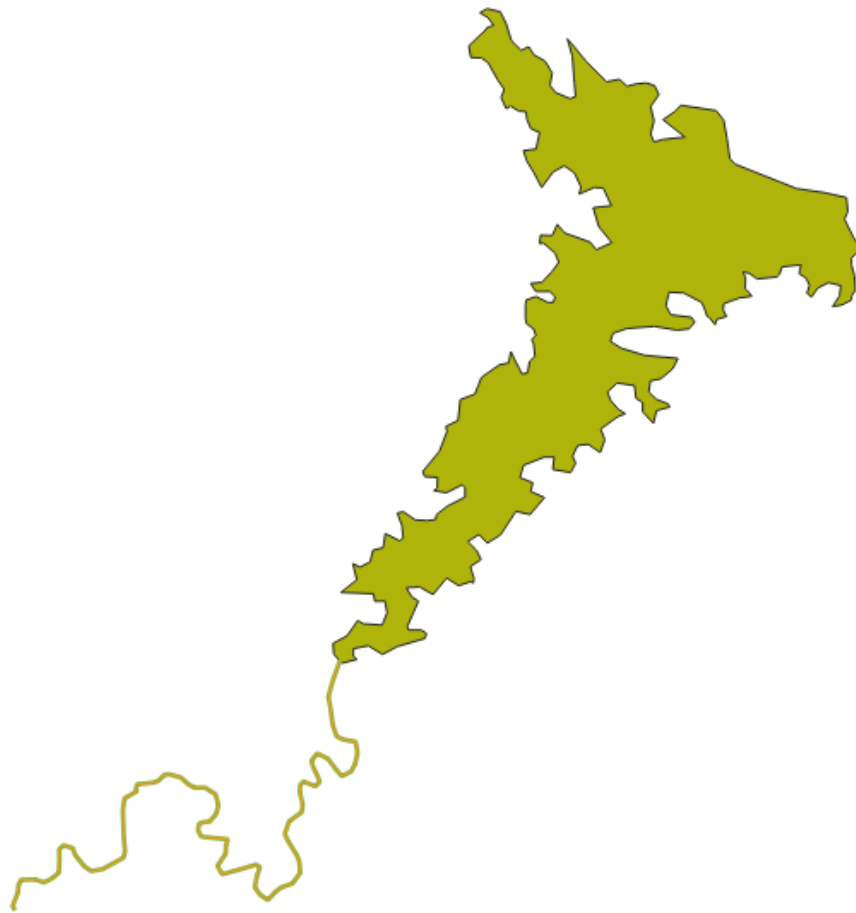
7. Export Shapefiles

- Right-click on polygon layer and line layer separately → Export → Save Features As.
- Format: ESRI Shapefile.
- Save to Location → OK.
- Output will generate shapefiles along with 5 supporting files (total 6 formats including **.shp**).

Results:

Extracted Shapefile.

Image 1



Conclusions:

This exercise demonstrated how to combine raster images into a virtual raster, generate false color composites, and digitize specific features.

- Water bodies were extracted as polygon shapefiles.
- Rivers were extracted as line shapefiles with symbology modifications.
- Exporting shapefiles created multiple associated files including .shp, .shx, .dbf, ensuring compatibility with GIS software.

Submitted by

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